The rearrangement of vinylcyclopropanes to cyclopentenes has proven to be a synthetically useful reaction (Hudlicky, et al. *Organic Reactions* **1985**, *33*). The mechanism of this rearrangement, however, is still under dispute (Davidson and Gajewski *J. Am. Chem. Soc.* **1997**, *119*, 10543; Houk, et al. *ibid*, 10545). Both concerted (orbital symmetry allowed) and stepwise rearrangements are theoretically possible.

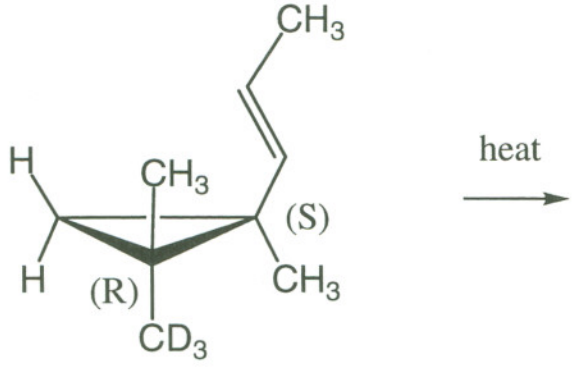
(a) Show both the concerted and stepwise pathways for rearrangement of the parent vinylcyclopropane **1**. For the concerted pathway, show the relevant orbitals and designate the stereochemical course of the reaction using the orbital symmetry description. For the stepwise pathway, use electron-pushing arrows to show how the bonds break and form.

(b) What would the expected rearrangement products of the isotopically labeled enantiomerically pure vinylcyclopropane **2** by both the concerted and stepwise pathways? Assume that only the more highly substituted cyclopropane bond breaks.

(c) Stereochemical markers are often used, as in part (b), to distinguish concerted from stepwise mechanisms. However, Carpenter has argued (*J. Am. Chem. Soc.* **1996**, *118,* 10329) that such a distinction may be invalid. How would you rationalize the formation of a single isomeric rearrangement product from vinylcyclopropane **2** if calculations indicated that the rearrangement occurred via a stepwise pathway (assume the calculations are accurate) ?

1

Organic



2